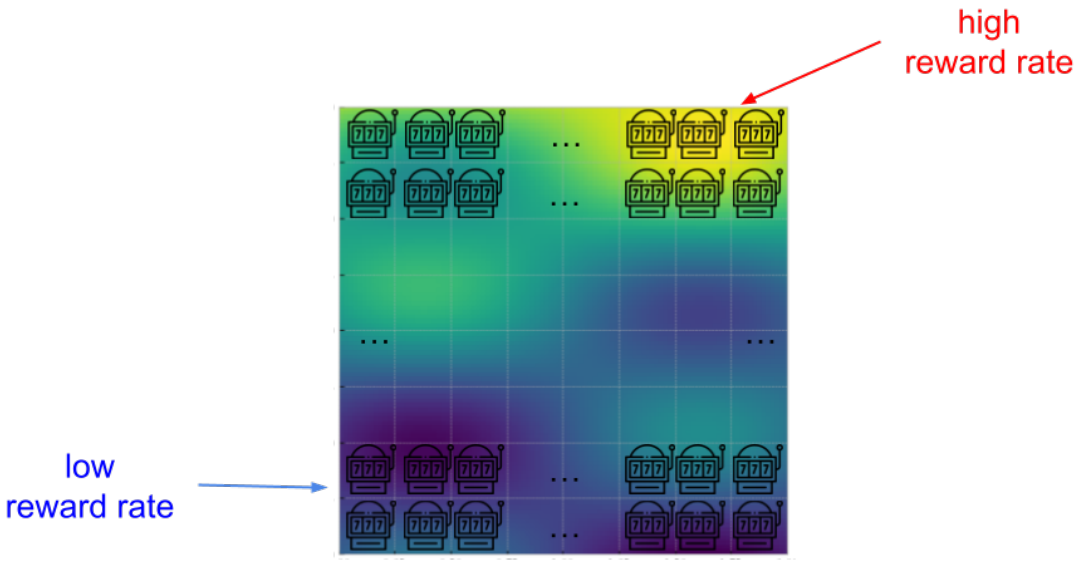


Figure 5.4 Bayesian optimization is similar to multi-armed bandit with infinitely many actions. Each infinitesimal point is a slot machine whose arm we could pull. Further, machines that are close to each other are correlated in the sense that they have similar reward rates.

Most formalizations of the MAB problem consider the binary setting, where a slot machine when pulled either returns a coin or doesn't return anything. The function value we may observe in BayesOpt, on the other hand, could take on any real value.

The main differences between MAB and BayesOpt are summarized in Table 5.1. Although these are fundamental differences, the tradeoff between exploration and exploitation in decision-making is present in both problems, so it's reasonable to aim to repurpose MAB policies to BayesOpt.

Table 5.1 The differences between multi-armed bandit and Bayesian optimization.

Criterion	Multi-armed bandit	Bayesian optimization
Objective to be maximized	Cumulative reward	Simple reward
Type of observation/reward	Binary	Real-valued
Number of actions	Finite	Infinite
Correlation between actions	No	Yes for similar actions

For the rest of this chapter, we will learn about two such policies, the motivation behind them, and how to implement them with BoTorch. We start with the Upper Confidence Bound policy in the next section.

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